

Chapter 9

Communication Function

9.1 Hardware.....118
9.2 Communication Parameters.....118
9.3 MODBUS Communication Protocol.....119
9.4 Write Parameters And Read Parameters.....123
9.5 Status Monitoring.....124
9.6 The Temporary Storage Function And Termporary Storage Address....125
9.7 Communication Wiring Definition.....125

9.1 Hardware

AC Servo Drive:

It has the serial communication function of RS-485 and it can realize parameter changes and status monitoring of servo system with MODBUS protocol.

9.2 Communication Parameters

Parameter	Name	Range	Default Value
PA-71	Drive ID No.	1~254	1

When using RS-485 communication, ID No. of the servo drive needs to be set to different values according to this parameter. The setting range of ID No.is from 1 to 254 and default value is 1. This ID No. represents the absolute address in communication network and if it is set repeatly, it will lead to abnormal communication.

Parameter	Name	Range	Default Value
PA-72	MODBUS Baud Rate	48~1152×100	96

You can choose RS-485 baud rates through this parameter and the baud rate that you chose should keep consistent with PC controller's.

Parameter Meaning：

If selection is 96×100，the baud rate is 9600.

Meanwhile, the communication protocol of RS-485 should be consistent with PC controller's

The setting values as below:

8，N，2（MODBUS，RTU）

"8" represents the transmission data is 8 bits. "N" represents that it doesn't use odd or even bits."2" represents the ending bit is 2.

Parameter	Name	Range	Default Value
PA-73	MODBUS Communication Protocol Selection	0~2	0

You can choose communication protocol of RS-485 through this parameter and the communication protocol should be consistent with PC controller's. The setting values as below:

0 : 8 , N, 2 (MODBUS , RTU)

1 : 8 , E, 1 (MODBUS , RTU)

2 : 8 , O,1 (MODBUS , RTU)

"8" represents the transmission data is 8 bits. "N,E,O" represents odd or even bit. While "N" represents it doesn't use this bit; "E" represents 1 even bit;"O" represents 1 odd bit;

"11"represents the ending bit is 1 and "2" represents the ending bit is 2.

9.3 MODBUS Communication Protocol

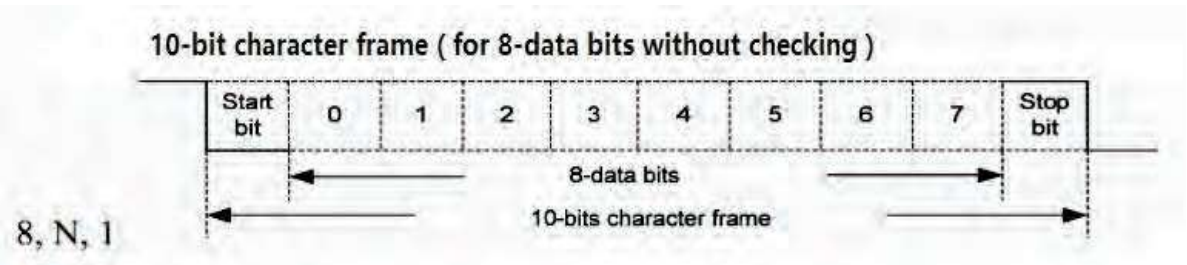
When using RS485 serial port communication, each servo drive must set servo drive ID in the parameter.

PC or controller should communicate with corresponding servo drive based on the ID No.And the baud rate need to set the parameters of the drive referring to the communicated parameters.Here, Modbus executes RTU(Remote Terminal Unit) mode.

Encoder Meaning

Each 8-bit dataes consists of two 4-bit hexadecimal characters. For example,1 byte data 64H.

Character Structure:



Communication Data Structure:

STX	The minimum time between the upper frame is 3.5 characters.
ADR	Communication address:1byte
CMD	Commmand code: 1byte
DATA(0)	Drive temperature is too highData: Nword=2Nbyte, N < =100

DATA(n-1)	
CRC	Checking code: 2byte
End1	The minimum time between the next frame is 3.5 characters.

The introduction for each iterm in the frame of communication data format is as following:

1、STX(Start of communication)

The minimum time between the upper frame is 3.5 characters.

2、ADR(Communication address)

The legal communication ranges from 1 to 254. Fro example, communicate with the drive of ID NO.16(hexadecimal10H):ADR=10H

3、CMD(Command code) And DATA(Data characters)

The data characters are depended on the command codes. The normal command codes are as follows:

(1)Command code 03H，read N words(16bit).

For example, 2 parameters are continuously read from the NO.5 parameter of the ID NO.01H servo drive.

ADR	01H
CMD	03H
The initial location of data	00H(High byte)
	05H(Lower byte)
Data bytes	00H(High byte)
	02H(Lower byte)
CRC Low	D4H(High byte)
CRC High	0AH(Lower byte)

ADR	01H
CMD	03H
Data bytes	04H
No.5 parameter	00H(High byte)
	96H(Lower byte)
No.6 parameter	00H(High byte)
	4BH(Lower byte)
CRC Low	5AH(High byte)
CRC High	28H(Lower byte)

(2) Command code 06H, Write 1 parameter.

For example, Write 100(0064H) into the No.5 parameter of the ID NO.01H.

Command:

ADR	01H
CMD	06H
The initial location of data	00H(High byte)
	05H(Lower byte)
Data bytes	00H(High byte)
	02H(Lower byte)
CRC Low	D4H(High byte)
CRC High	0AH(Lower byte)

Response:

ADR	01H
CMD	06H
The initial location of data	00H(High byte)
	05H(Lower byte)
Data bytes	00H(High byte)
	02H(Lower byte)
CRC Low	D4H(High byte)
CRC High	0AH(Lower byte)

4、CRC Frame Checking Calculation:

The procedures of checking calculation:

Step 1: Initializes a 16-bit register with FFFFH, which named as CRC register.

Step 2: XOR(exclusive or data processing) is calculated between the first character of command information and the lower byte of 16-bit CRC register and then the result is stored in the CRC register.

Step 3: Check the lowest bit(LSB) of the CRC register. If it is 0, then move to the right one.

If it is 1, then move to the right one and after that, XOR(exclusive or data processing) operation is calculated with A001H.

Step 4: Back to step 3 and until the operation of step 3 has been executed for 8 times, you can enter into step 5.

Step 5: The next byte of command information is operated from step 2 to step 4 until all bytes have been processed completely as above, then the information of CRC register is CRC frame checking.

Note:

After calculatinf the CRC frame checking, must firstly fill in the lower bits of CRC and then fill in the high bits in the command information.

4、CRC Frame Checking Calculation:

The procedures of checking calculation:

Step 1: Initializes a 16-bit register with FFFFH, which named as CRC register.

Step 2: XOR(exclusive or data processing) is calculated between the first character of command information and the lower byte of 16-bit CRC register and then the result is stored in the CRC register.

Step 3: Check the lowest bit(LSB) of the CRC register. If it is 0, then move to the right one.

If it is 1, then move to the right one and after that, XOR(exclusive or data processing) operation is calculated with A001H.

Step 4: Back to step 3 and until the operation of step 3 has been executed for 8 times, you can enter into step 5.

Step 5: The next byte of command information is operated from step 2 to step 4 until all bytes have been processed completely as above, then the information of CRC register is CRC frame checking.

Note: After calculatinf the CRC frame checking, must firstly fill in the lower bits of CRC and then fill in the high bits in the command information.

For example, Continously to read 2 parameters in the NO.5 paramter of the ID NO.01H servo drive. From ADR to the last byte of data, the final information of CRC register that was calculated is 0AD4H.Then the command information is as follows:

Note: ByteD4H should be served before the byte0AH.

ADR	01H
CMD	03H
The initial location of data	00H(High byte)
	05H(Lower byte)
Data bytes	00H(High byte)
	02H(Lower byte)
CRC Low	D4H(High byte)
CRC High	0AH(Lower byte)

5、Ending with End1 communication:

The minimum time between the next frame is 3.5 characters.

9.4 Write Parameters And Read Parameters

1、Write PA group parameters

PA group parameters of the servo drive refer to the chapter 6 of the servo manual.

Each parameter is represented by 16 bits and their communication address is determined by the parameter No..For example,parameter 1(PA-0) is 0X0000; Parameter 2(PA-1) is 0X0001 and other parameters are like this.

2、Write P3 group parameters

P3 group parameters of the servo drive refer to the chapter 6.2 of the servo manual.

Each parameter is represented by 16 bits and their communication address is determined by the parameter No..For example, parameter 1 (P3-0) is 0X0100H; Parameter 16(P3-15) is 0X010FH. Other parameters are deduced in turn.

3、Write P4 group parameters

P4 group parameters of the servo drive refer to the chapter 6.3 of the servo manual.

Each parameter is represented by 16 bits and their communication address is determined by the parameter No..For example, parameter 1 (P4-0) is 0X0200H; Parameter 16(P3-15) is 0X020FH. Other parameters are deduced in turn.

4、The format of writing parameters and reading parameters

The introduction of the format of writing parameters and reading parameters(state volume reading refers to chapter 1.5):

The parameter must be a decimal integer. The values of parameters with decimal points displayed on the drive are amplified in the proccess of reading and writing, which leads to make them converted into decimal integer.

PA Group Parameters	Drive Display	Communication Operation	Transformed Mode
1	315	315	No
63	1.00	100	Magnify 100 times
57	0100(binary)	4(decimal)	Binary to decimal

The values of parameters in t he manual can be read and wrote through communication.

The details refer to coresponding introductions of parameters in the manual.

9.5 Status Monitoring

The internal states in the servo drive can be read through the terminal of RS485 communication, but it can not be wrote. The states are saved as 16-bit data. And when values of parameters with decimal points are read by the communication terminal, they are amplified to 10 times or 100 times. This is like as the parameter reading. The order of status are as follows:

1000H	Display motor speed
1001H	Display the current position is 5-bit low
1002H	Display the current position is 5-bit high
1003H	Display position command(command pulse accumulation) is 5-bit low.
1004H	Display position command(command pulse accumulation) is 5-bit high.
1005H	Display position deviation is 5-bit low.
1006H	Display position deviation is 5-bit high .
1007H	Display motor torque
1008H	Display motor current
1009H	Display control mode
100AH	Display temperature
100BH	Display speed command
100CH	Display torque command
100DH	Display absolute position of the rotor in a roll is 5-bit low
100EH	Display absolute position of the rotor in a roll is 5-bit high .
100FH	Display input terminal state
1010H	Display output terminal state
1011H	Display encoder input signal
1012H	Display voltage value of main line of main circuit
1013H	Display alarm code
1014H	Display logic chip version number

9.6 The Temporary Storage Function And Termporary Storage Address

According to the needs of customers, it is necessary to update the parameter values continuously in the process of running the drive. In order to ensure the life of EEPROM and speed up the efficiency of program execution, the temporary storage function of parameters is added. When the parameter is modified with the corresponding temporary address, the parameter can be modified, but it can not be saved. When the drive is repowered on, the parameter will restore to the initial value. For example:

The address of the forward torque limit value is ensured to be 0x0022, when the parameter is modified to 200 with the address , it is saved in the EEPROM. The parameter is still 200 without lost after the drive is powered off. While the address of the temporary forward torque limit value is 0x00A2,when the parameter is modified to 200 with the address , the value is set in and will function immediately, but the initial value will be restored to 300 after repowering on.

The communication address of the saved parameters of the PA group parameters is described in the charpter of 1.4 and the parameter description part of the drive manual. The temporary communication addresses of PA group parameters are offset 0x0080 than the addresses of storaged parameters, for example:

(1) The saved parameter communication address of the forward torque limit value is 0x0022, and the temporary parameter communication address of forward torque limit value is 0x00A 2 after offsetting 0x0080.

(2) The saved parameter communication address of the reverse torque limit value is 0x0023.After offsetting 0x0080, the temporary parameter communiction address of the reverse torque limit value is 0x00A3.

9.7 Communication Wiring Definition

Rs485 Communication Signal		
Pin	Definition	Picture
4	Rs485-	Connect with No.4 and No.5 pin 
5	Rs485+	

WARRANTY

❶ The company solemnly promises that users will enjoy the following warranty services from the date of purchase of products from our company (hereinafter referred to as the manufacturer).

❷ Since the product was purchased by the user from the manufacturer, enjoy the following three guarantee services:

- ⌑ Return, replacement and repair within 30 days of delivery:
- ⌑ Replacement and repair within 90 days of delivery:
- ⌑ Repair within 18 months of delivery:
- ⌑ Except when exporting abroad.

❸ This product enjoys lifetime paid service from the date of purchase by the user from the manufacturer.

❹ Disclaimer: Product failure caused by the following reasons is not covered by the manufacturer's free warranty service:

- ⌑ Failure caused by the user's use and operation in accordance with the requirements of the «Instruction Manual»:
- ⌑ Failure caused by the user to repair or modify the product without communicating with the manufacturer:
- ⌑ Failure caused by abnormal aging of the product due to poor user environment:
- ⌑ Failures caused by natural disasters such as earthquakes, fires, floods or abnormal voltages:
- ⌑ Damage to the product during transportation (the transportation method is specified by the customer, and the company assists in handling the cargo consignment procedures)

❺ Under the following conditions, manufacturers have the right not to provide warranty services:

- ⌑ When the manufacturer's product logo, trademark, nameplate, etc. are damaged or unrecognizable:
- ⌑ When the user fails to pay the purchase price in accordance with the signed contract:
- ⌑ The user intentionally conceals the manufacturer's after-sales service unit when the product is installed, wired, operated, maintained or otherwise improperly used

❻ For the service of return, replacement and repair, the company must return or return to the company, and it can only be returned or repaired after confirming the responsibility vested.